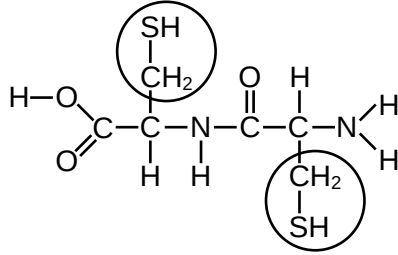


2(a)	either R group circled ; 	1
2(b)(i)	disulfide (bonds) ;	1
2(b)(ii)	<u>exocytosis</u> ; <i>plus any two from:</i> either: (idea of) vesicles will be large enough to contain many mucins / ref. to bulk transport ; vesicles forming from Golgi, body / apparatus ; vesicles moved by microtubules / cytoskeleton (to cell surface membrane) ; vesicles fuse with <u>cell surface membrane</u> ; active process / requires ATP ; AVP ; e.g. mucins are, polar / hydrophilic, so cannot cross, phospholipid bilayer / hydrophobic core of membrane or: mucin molecules are hydrophilic ; can exit via protein channels ; by facilitated diffusion ;	3
2(c)	<i>any two from:</i> (because mucus is too thick) cilia, have difficulty / AW, moving the mucus (upwards) ; pathogens, build up / not removed / AW ; more chance of, infection / disease ; AVP ;	2

Question	Answer	Marks
2(d)(i)	deletion ;	1
2(d)(ii)	transcribed / template ; R transcription strand I anti-sense / coding, strand	1
2(d)(iii)	AUC AUU GGU GUU ;	1
2(d)(iv)	any three from: 1 idea of 3 bases coding for 1 amino acid ; 2 introns removed from, primary transcript RNA / AW ; A gene splicing / RNA splicing A mRNA does not contain introns R if not in correct context (primary transcript to mRNA) 3 introns / non-coding DNA, do not code for amino acids ; 4 (only) exons are joined together (to form mRNA) ; 5 DNA triplet / mRNA codon, for STOP does not code for an amino acid ; 6 methionine at start / first amino acid / amino acid coded for by START codon, removed ; 7 AVP ; e.g. ref. to (upstream) enhancer sequences ref. to (downstream) terminator sequences ref. to (non-coding) regulatory sequences / promoter	3